



Clinical Practice Guideline

Parenteral Nutrition

Scope

Site	Service/Department/Unit	Disciplines
SCGH	Hospital-Wide	All Staff,

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Introduction

Total Parenteral Nutrition (TPN) refers to the provision of nutrition via a central vein¹. Peripheral Parenteral Nutrition (PPN) is similar to TPN, however contains a lower osmolality and concentration of nutrients allowing for administration via a peripheral vein.²

This guideline outlines the procedures and responsibilities of individuals and the health service organisation in relation to the safe administration and management of Parenteral Nutrition (PN). It ensures the delivery of a uniform, integrated multidisciplinary approach with management that is consistent with best practice and in accordance with existing SCGOPHCG clinical, nursing, medication and infection control policies and guidelines.

Risk Statement

Non-compliance with this policy will:

Breach legislative requirements	☐	Impact on Patient Quality of Care	☒
Breach National/State/Hospital Policy	☒	Impact on Patient Safety	☒
Breach professional standards	☒	Misconduct	☐
Breach SCGOPHCG Mission & Values	☐	Staff Safety	☐
Other:	☐		

Definitions

Central Venous Catheter (CVC)	CVC is a venous catheter that is inserted via the internal or external jugular, subclavian, axillary, or femoral veins and advanced until the tip is located in the vena cava.
Enteral Nutrition (EN)	Delivery of a specialised liquid food mixture containing protein, carbohydrates, fats, vitamin and minerals directly into the gastrointestinal tract via a tube, catheter or stoma that bypasses the oral cavity.
Extravasation	The unintentional instillation or leakage of a medication or substance out of the blood vessel and into the surrounding tissue.
Home Parenteral Nutrition (HPN)	Administration of PN by a patient or carer in the home environment.
Peripherally Inserted Central Catheter (PICC)	A PICC is a venous catheter that is inserted via the brachial, basilic or cephalic veins in the upper arm and advanced until the tip is located in the superior vena cava.
Peripheral Parenteral Nutrition (PPN)	Short-term intravenous (IV) infusion of dextrose, amino acids, lipids and electrolytes via a <u>peripheral vein</u> in patients who are unable to assimilate adequate nutrition via the enteral route.
Phlebitis	The presence of inflammation within a vein and is clinically associated with pain, tenderness, induration and erythema along the course of the superficial vein.
Parenteral Nutrition Associated Liver Disease (PNALD)	Liver dysfunction caused by intestinal failure, or inability to digest and absorb nutrients enterally, that occurs in the setting of PN use.
Total Parenteral Nutrition (TPN)	IV infusion of dextrose, amino acids, lipids, micronutrients and electrolytes through a <u>central vein</u> to patients who are unable to assimilate adequate nutrition via the enteral route.
Triple-Chamber Bag (TCB)	Standardised, commercially available, all-in-one parenteral nutrition admixture containing lipid, glucose and amino acid solutions in three separate chambers.

1. Indications for Parenteral Nutrition

PN is indicated when the Gastro Intestinal Tract (GIT) is not functional, suitable for use or accessible, or when a patient's nutritional requirements cannot be met adequately by oral or enteral nutrition (EN).³ PN may be required in the following scenarios:^{1, 4, 5}

- Complete bowel obstruction.
- Suspected gut ischaemia.
- Gastrointestinal fistula or high output enterocutaneous fistula (>500mL per day).
- High risk of upper GIT bleeding (e.g., oesophageal varices or low platelet count).
- Severe persistent diarrhoea or malabsorption.
- Chyle leak (>1000mL / day or leak unable to be managed with oral or enteral nutrition (EN)).
- Congenital abnormalities of the GIT.
- High risk surgical anastomoses or unrepaired anastomotic leak (if distal feeding not possible).
- Medical or surgical ileus
- High output stoma (> 1.2L stomal output per day) despite dietary and medication management strategies.
- Gut rest during chemotherapy (in some cases).⁶
- Inability to obtain enteral access (e.g., anatomical abnormalities, repeated failed nasogastric tube (NGT) insertion attempts, severe facial injuries).

PPN at SCGH is only available on the Intensive Care Unit (ICU) and under General Surgery, Gastroenterology and Transplant specialties on wards G61, G63 and G71.

If PN is deemed necessary, medical staff need to determine whether TPN or PPN is a more appropriate form of nutrition support (refer to Table 1 below).

Table 1: Additional indications for TPN or PPN

Indications for TPN	Indications for PPN
<u>Patients on the ward:</u> • The patient is well nourished and PN is likely to be required for more than 5 days. • The patient is malnourished or at high nutrition risk and enteral / oral access is unavailable or inadequate. • The patient has been receiving PPN for 5 – 7 days and is unable to be transitioned to oral or enteral nutrition.	• Supplemental nutrition via a parenteral route is required for a short duration (ideally less than 5-7 days).^{2, 3, 7} • Central access for TPN is unavailable or unsuitable. • Awaiting access to enteral feeding or if enteral access is continuously lost. • Malnourished patients where PPN can be used as a 'bridging' step to longer term forms of nutrition support (e.g., EN or TPN) or as a supplementary nutrition source if the patient is not tolerating full EN.⁷
<u>Patients in the ICU:</u> • The patient is well nourished on admission to ICU or at low nutrition risk, and oral / enteral nutrition is contraindicated for more than 3 – 7 days.⁴ • The patient is malnourished on admission to ICU or at high nutrition risk, and oral / enteral nutrition is contraindicated for more than 24 – 48 hours.^{3, 4} • Supplementary TPN can be used when patients relying on EN for full nutrition support have been unable to achieve more than 60 - 80% of nutritional targets despite all strategies to maximise EN tolerance.⁴	

2. Contraindications for Parenteral Nutrition

Both forms of PN are contraindicated in the following situations:^{4, 8}

- The patient has a functional, accessible GIT and can meet more than 80% of nutritional requirements via oral or enteral intake.^{4, 8}
- End of life care.
- Severe hyperglycaemia.
- Severe hyperlipidaemia.
- General contraindications for administering an intravenous infusion.
- Congenital abnormalities of amino acid metabolism.
- Severe blood coagulation disorders.
- Known severe hypersensitivity to egg, soya proteins, corn (maize) and corn products, peanut protein, components of the container, or to any other ingredient in the feeding solution.

Additional contraindications specific to the different modes of PN are outlined in Table 2 below.

Table 2: Additional contraindications for TPN or PPN

Contraindications for TPN	Contraindications for PPN
• Patient is well-nourished or at low nutrition risk, and likely to commence oral / enteral intake in less than 5 – 7 days if on the ward or less than 3 days if in ICU.^{3, 4} • Severe renal insufficiency without the possibility of dialysis or haemofiltration. • Unstable conditions (e.g., hyperosmolar coma).	• Patient is well-nourished and likely to commence oral or enteral intake in less than 5 days.³ • Fluid restriction.¹ • Lymphoedema.⁷ • Renal failure (PPN contains set quantities of electrolytes that may have inadequate clearance). • Difficult IV vascular access or predicted poor vascular access.⁷ • The patient is not located on G61, G63, G71 or ICU.

3. Total Parenteral Nutrition

TPN at SCGH is prescribed by the TPN Team, who are a referral-based consultancy service based out of the ICU. The TPN Team includes an intensivist, senior dietitian and senior pharmacist.

All ward patients meeting the inclusion criteria for TPN require an e-referral to the TPN Team from the treating team and should be referred to the ward dietitian for a comprehensive nutrition assessment prior to review by the TPN Team.

If TPN is required to commence the same day, the treating team should also notify the TPN Team by contacting the ICU (extension 71010) or paging the TPN pharmacist (pager 3893). Any referral received after 12.00 hours will be seen on the following day. Patients in ICU do not require an e-referral and can be commenced on TPN by any intensivist.

The role of the treating team is to:

- Complete the e-referral
- Organise central vascular access
- Order baseline bloods (urea and electrolytes (U&Es), liver function tests (LFTs), magnesium, calcium, phosphate),
- Administer the prescribed TPN formulation.
- Monitor overall fluid input and prescribe additional IV hydration if required
- Continue to provide all usual medical and nursing care

The role of the TPN Team is to:

- Assess the patient's suitability for TPN
- Prescribe daily TPN formula and supplemental electrolytes
- Order ongoing PN-related pathology
- Monitor the patient for any TPN-related complications

3.1 Central Venous Access

- TPN solutions have a high osmolality and are therefore only suitable for delivery via a central line.⁹ TPN must NOT be administered via peripheral vein.
- TPN may be delivered via Central Venous Catheter (CVC) or a Peripherally Inserted Central Catheter (PICC). TPN may also be delivered via a tunnelled central line such as a Hickman® or Broviac® catheter or an implanted port if available.⁹
- Refer to Non-Tunnelled Central Venous Catheter and Peripherally Inserted Central Catheter Clinical Practice Guideline (SCGOPHCG) for information regarding vascular pathway for insertion of a CVC / PICC.
- TPN should be infused via a dedicated lumen in the central line. For this reason, when referring a patient to SCGH IV Access Service or Interventional Radiology for PICC insertion, a double lumen PICC should be requested.
- Documentation of CVC / PICC insertion details should be completed in the SCGH Intravascular Device Insertion Record Central Vascular Access Device (MR864.2) at the time of insertion.

3.2 TPN Formulation

- Pharmacy stocks several different Triple-Chamber Bags (TCBs) for central venous administration, including Olimel® N9-840, Olimel® N9-840E, Olimel® N12-640E and SMOFKabiven® in varying volumes. Refer to [Appendix 2](#) for further information.
- In some situations, a patients' nutritional requirements will not be met with one of the available TCBs. These situations may include, but are not limited to, patients on a tight fluid restriction, concurrent propofol infusion or high serum triglycerides levels. These patients may be prescribed a customised TPN bag using a combination of individual commercially manufactured base solutions. See [Appendix 3](#) for further information.
- All custom TPN bags and additions made to existing TCBs are compounded by Aseptic Pharmacy Services in a designated compounding area under a sterile laminar flow hood.
- The amount of glucose in TPN should not exceed 5 g/kg/min⁴.
- Intravenous lipid, including non-nutritional lipid sources, should not exceed 2.0 g lipids / kg / day (or 1.5g lipids / kg / day for critically ill patients) and should be adapted to individual tolerance⁴.

3.2.1 Electrolytes

- Each TPN TCB contains a pre-set number of electrolytes.
- Additional electrolytes can be ordered by the TPN Team on the TPN Prescription Order and will be added to the TPN bags by Aseptic Pharmacy Services.
- Due to stability issues, there is a limited amount of phosphate, magnesium and calcium that can be added to the TPN bags therefore separate IV supplementation is often required if serum levels are very low.
- The maximum allowable electrolyte additions for each TPN bag are outlined in [Appendix 2](#).

3.2.2 Vitamins and Minerals

- All patients on TPN should receive a daily dose of vitamins and minerals / trace elements.⁴
- Vitamins and trace elements are automatically added to TPN bags by Aseptic Pharmacy Services unless requested otherwise by the TPN Team.
- Refer to [Appendix 3](#) for available micronutrient solutions,

3.2.3 Insulin

- Actrapid® may be added to TPN bags in Aseptic Pharmacy Services for patients with consistently elevated blood glucose levels (BGLs).
- The amount of Actrapid® added to the bag will depend on the patient's usual insulin regime, current BGLs and amount of insulin received by the patient in the preceding 24 hours.
- If the patient is already under the care of the Endocrinology team, the TPN Team will liaise with the Endocrinology team prior to adjusting the insulin level in the TPN bag.
- If the patient is in the ICU, a separate insulin infusion may be prescribed and administered using a 3-way tap.³

3.3 TPN Prescribing and Ordering

- TPN is prescribed daily by the TPN Team using TPN Prescription Order.
- All new referrals for TPN will receive a comprehensive nutrition assessment by the TPN dietitian, including risk of refeeding syndrome, nutritional targets, and 'goal' PN formulation.
- Patients who are identified at risk of refeeding syndrome should be prescribed 100-300mg thiamine intramuscularly or intravenously at least 30 minutes prior to initiation of TPN, and then daily for the first 5 – 10 days of feeding. TPN should be commenced slowly and increased gradually, whilst monitoring for electrolyte imbalances.¹⁰ Refer to [Refeeding Syndrome Guideline](#) for more information.
- Orders must be signed by the intensivist and sent to Aseptic Pharmacy Services via the internal chute by 12.00 hours. Aseptic Pharmacy Services require advance notification if any orders are likely to be delayed.
- TPN bags are delivered to ward areas between 16.00 hours and 18.00 hours each day and should be refrigerated until ready for use.
- In most cases, referrals for TPN that are received after 12.00 hours will be reviewed by the TPN Team during the TPN ward round the following day.
- If TPN is required urgently, after the 12.00 hours deadline for Aseptic Pharmacy Services, a standard 1000mL TCB (Olimel® N9-840) from ICU may be prescribed on the directive of the TPN Team/intensivist. In this case, the TPN Team will prescribe the TPN on the Intravenous Fluid Therapy Orders (MR410/825), and provide the TPN bag, a 1.2 micron in line filter and a light protective covering. The TPN bag and light protective covering must be labelled, signed and dated as for all other infusions.
- For ward patients, if appropriate, the TPN Team will prescribe a sliding scale insulin regimen on the Insulin Subcutaneous Order and Blood Glucose Record – Adult (MR846) chart at the commencement of TPN therapy.

3.4 TPN Regimens

- TPN commencement and change time is 18.00 hours.
- When TPN is initiated, it is usually administered continuously over a 24 hour period.
- Once the patient is clinically stable, the TPN Team may transition the patient to an intermittent or overnight TPN regimen to allow the patient periods of disconnection.
- Patients on intermittent or overnight TPN will have a maximum infusion rate, total infusion volume, ramp duration, and taper duration outlined on the TPN Prescription Order. The tapering rate helps to reduce the risk of hypoglycaemia when the TPN is ceased.

4. Peripheral Parenteral Nutrition

PPN may be indicated for short term nutritional support (ideally 5 – 7 days²) for appropriate patients, who have adequate peripheral venous access, and where central access is unavailable or unsuitable. PPN has a lower osmolality (760 mOsmol/L) than TPN, which allows it to be safely administered via a peripheral vein.⁷

PPN prescription is managed by the patients' treating team, in conjunction with Dietetics and Pharmacy.

PPN at SCGH is only available on the Intensive Care Unit (ICU) and under General Surgery, Gastroenterology and Transplant specialties on wards G61, G63 and G71.

4.1 Peripheral Venous Access

- PPN must only be administered via a peripheral line. In certain circumstances, upon Consultant approval, PPN can be administered via a central line if central access is already in situ.
- PPN must be delivered via a dedicated 20-22G Peripheral Intravenous Vascular Catheter (PIVC) into the largest peripheral vein available in the forearm to minimise phlebitis and allow for adequate blood flow around the device (avoid the cubital fossa).^{1, 7, 11, 12} If a midline is already in situ, PPN can be delivered via a midline instead of a PIVC. However, a midline is NOT to be inserted for the purpose of PPN only. If a patient progresses onto requiring TPN, a midline must NOT be used for the administration of TPN.
- Avoid delivering PPN via PIVC in the lower limb or hand.
- All PVCs must have a j-loop attached.
- Aseptic technique is to be maintained at all times during insertion and maintenance of the PVC.
- Documentation of the PIVC insertion details are to be completed on the Peripheral Intravenous Vascular Catheter Insertion Record (MR864.2).
- All PVCs are to be re-sited at 72 hours unless otherwise clinically indicated. Refer to Peripheral Intravenous Catheters (PIVC) - Insertion, Maintenance and Removal (SCGOPHCG).

4.2 PPN Formulation

- PeriOlimel® N4-600E is the only PPN formulation available for peripheral administration. It is a 2000mL TCB.
- Due to its lower osmolality (760 mOsmol/L), the solution can be administered via either a peripheral vein or a central vein.⁷

**Note: The TPN bags (Olimel® N9-840E, Olimel® N12-640E, and SMOFKabiven®) are hypertonic and must NOT be administered peripherally.*

4.2.1 Electrolytes

- PeriOlimel® N4-600E contains a pre-set number of electrolytes.
- Additional electrolytes cannot be added to the bag.
- If additional supplementation is required, supplement via a separate peripheral cannula. The responsibility for replacement of any electrolytes is with the treating team.

4.2.2 Vitamins and Minerals

- PeriOlimel® N4-600E does not contain any vitamins or trace elements.
- If daily vitamins and trace elements are required, these can be charted and administered separately (externally to the PPN bag) via a separate peripheral cannula.

4.3 PPN Prescribing and Ordering

- PPN must be prescribed by the treating team (General Surgery, Gastroenterology or Intensive Care specialties) in conjunction with Dietetics and Pharmacy. If the dietitian is unavailable, PPN may still be prescribed by the treating team, if appropriate, until the dietitian is available to review the patient.

- PPN Order Form (MR526) to be completed and signed by the treating team medical officer.
- Dietitian to complete the 'Nutritional Requirements' section of the PPN Order Form (MR526) and document the recommended starting and target rate of PPN.
 - Patients not at risk of refeeding syndrome can be commenced at the intended final infusion rate, which would normally be 80 mL/hr.⁷
 - The risk of refeeding is much less in those receiving PPN compared with TPN, due to the lower glucose content of the PPN formulation.
 - For patients at risk of refeeding syndrome, refer to the Refeeding Syndrome Guideline and commence PPN slowly (a suggestion is commencing at 50% of the intended final infusion rate).
- Treating team or ward pharmacist / dietitian to fax order form to inpatient Pharmacy by 15.00 hours for same day delivery.
- For PPN orders received by Pharmacy after 15.00 hours, PPN bag will be supplied the following day for an 18.00 hour administration time. PPN is not considered an 'urgent treatment'.
- PeriOlimel® N4-600E PPN bag will be delivered to the ward labelled with the patients' name and the directions: "For peripheral parenteral nutrition use only."
- Store PPN bag at room temperature on ward.

4.4 PPN Regimens

- PPN is designed to be administered over 24 hours at the rate determined by the dietitian on the PPN Order Form (MR526).
- PPN commencement and change time is 18.00 hours. Discard PPN after 24 hours.
- The dietitian and / or treating team will make the decision regarding the continuation or cessation of PPN prior to the next scheduled bag change.

5. Parenteral Nutrition Administration

- Observe the five moments for hand hygiene. Follow aseptic technique at all times and use the risk assessment outlined in Infection Prevention and Management Aseptic Technique Policy (SCGOPHCG). This must include use of appropriate Personal Protective Equipment (PPE) to prevent contamination and mucosal or conjunctival splash injuries.
- PN is to be administered through an infusion pump at the rate prescribed on the PN prescription.
- An in-line filter infusion line should be used to prevent complications from particulate matter, micro precipitates or micro-organisms that may be present in the PN solution.¹³ It should be replaced routinely with the change of PN bags and giving sets every 24 hours.
- PN infusion must be commenced at least 24 hours prior to expiry date and time on PN bag label and commenced / changed every 24 hours at 1800 hours (even if volume remains). Discard any remaining PN after 24 hours into clinical waste bin.
- Prior to commencing or changing PN infusions the patient identification must be checked by three indicators (e.g., name, date of birth, address or UMRN) SCGH Hospital Policy Patient Identification and Procedure Matching.
- **PN must be checked at the bedside by two nurses**, one of whom is a Registered Nurse (RN).
- Both nurses must document administration by signing PN prescription order form. When current PN bag is disconnected from patient, the same prescription form is to be signed by two nurses to indicate date and time infusion completed.
- Protect PN bag from light with protective covering provided by Pharmacy.
- Check labelled PN bag against PN prescription.
- The route of administration must be identified on all administration lines and include the date and time that the line is commenced.

- Use only if the bag is not damaged and inspect for uniformity of colour and absence of precipitates.
- If using a PPN bag or standard TPN TCB from ICU imprest, ensure the bag is mixed well/activated prior to attaching the PN bag to the dedicated peripheral cannula or CVC lumen (see Figure 1.0). The solution will come in a vacuum sealed bag* with slits at the top and bottom of the bag to assist with opening. DO NOT use scissors to open bag as internal PN solution bag may become damaged.

**Note: The vacuum sealed bag uses the Oxydetect™ system which offers visual proof that oxygen has not infiltrated bag.*

Light yellow = safe to use.

Blue = DO NOT USE.

- Do not introduce any additional medications to the PN infusion bag or administer any bolus IV medications via the lumen dedicated to PN therapy.
- Do not co-infuse blood transfusions with PN.
- Insulin is the only medication which can be co-infused with TPN via the dedicated TPN lumen (in ICU patients only). Connect new insulin infusion line to TPN infusion line using 3 way tap, only when TPN is changed at 18.00 hours. Until this time the insulin must be infused using a separate IV line.³
- Refer to [Appendix 5](#) for more information regarding trouble-shooting common problems that may arise with inpatient PN.

To mix / activate PPN bag or standard TPN TCB from ICU imprest:

1. Ensure product is at room temperature.
2. Lie TPN solution bag flat on surface at waist height.
3. Manually roll the bag onto itself, starting at the top of the bag (hanger end). The non-permanent seals will disappear from the side near the inlets.
4. Continue to roll until the seals are open along approximately half of their length.
5. Mix by inverting the bag at least 3 times.
6. After reconstitution, the mixture is a homogenous emulsion with a milky appearance.



Source: Baxter Healthcare Ltd

REQUIREMENTS

- TPN solution:
 - PPN solution, Infusomat® Space line SafeSet and 1.2 micron filter

OR

 - TPN solution with pre primed giving set and attached 1.2 micron filter

OR

 - TPN solution from ICU imprest, Infusomat® Space line SafeSet and 1.2-micron filter
- Current valid PN prescription
- 10 mL sodium chloride 0.9% Posiflush SP™ Syringe
- Non-sterile gloves
- Large 70% alcohol wipe
- 3-way tap (ICU only)

PROCEDURE

1. Perform hand hygiene.
 2. Clean trolley with combined detergent disinfectant wipe. Allow to dry.
 3. Gather equipment for procedure.
 4. Perform hand hygiene.
 5. If using PPN or a standard TPN TCB from ICU imprest, complete the following:
 - a. Mix / activate the PN solution (refer to Figure 1.0).
 - b. Attach 1.2-micron filter to end of giving set and connect to PN bag. In ICU, attach 3-way tap to distal end of giving set.
 - c. Prime the PN line using Infusomat® Space line SafeSet.
 6. Don non-sterile gloves. Protect key parts using a non-touch technique. Scrub the end of the needleless access port or positive pressure valve with a large 70% alcohol wipe for 30 seconds using different areas of the wipe. Allow to air dry for 30 seconds.
 7. Flush the lumen with sodium chloride 0.9% Posiflush SP™ Syringe, using a pulsatile motion (push-pause).
 8. Connect giving set (with 1.2-micron filter attached).
 9. Ensure PN bag is covered by light protective overpouch supplied by Pharmacy.
 10. Commence infusion at prescribed rate.
 11. Dispose of waste appropriately. Remove gloves.
 12. Perform hand hygiene.
 13. Clean trolley with combined detergent disinfectant wipe. Allow to dry.
 14. Perform hand hygiene.
- Refer to [Appendix 4](#) for pump programming guideline.

6. Monitoring and Management of Parenteral Nutrition**6.1. Monitoring and Management of CVC/PICC**

- Assess insertion site and associated limb at least once per nursing shift, prior to any access, at dressing changes or as clinically indicated.
- Monitor for signs of redness, swelling or pain, leaking, occlusion or migration.
- If infected, central lines need to be removed, the infection treated with antibiotics and the central line not replaced until the infection is controlled.¹⁴
- Refer to Non-Tunnelled Central Venous Catheter and Peripherally Inserted Central Catheters (SCGOPHCG) and Non-Tunnelled and Tunnelled Large Bore Central Venous Catheters (SCGOPHCG).

6.2. Monitoring and Management of PIVC

- Observe PIVC insertion site for signs of phlebitis, leakage, occlusion, infiltration or extravasation prior to use and each nursing shift.
- Document inspection of the insertion site each 8 hours on the PIVC insertion record (MR864.2), by following the Peripheral Intravenous Assessment Score (PIVAS).

If extravasation is suspected:

- Perform hand hygiene and apply PPE.
- Stop infusion immediately.
- Attempt to aspirate any infusion fluid from the PVC, using a sterile 10 mL syringe.
- Remove PIVC.
- Dispose of equipment appropriately, remove PPE and perform hand hygiene.
- Notify medical officer for review / assessment of PIVC site.
- Document incident in the patient's medical notes and report using the Clinical Incident Management System (CIMS).

Refer to Peripheral Intravenous Catheters (PIVC) - Insertion, Maintenance and Removal of (SCGOPHCG).

6.3. Monitoring PN Infusion / Clinical Status

- Monitor and record the patient's core physiological observations a minimum of 4 hourly for the duration of the PN infusion.
- Weigh patient prior to commencement of PN and then repeat twice weekly or as clinically indicated.
- Record fluid intake and output on ICU Flow Chart (MR116), or Fluid Chart (MR904) and Fluid Balance Summary (MR873).
- Establish baseline BGL prior to commencement of PN and record on the Insulin Subcutaneous Order and Blood Glucose Record – Adult (MR846) chart or ICU Flow Chart (MR116). Monitor BGLs 6 hourly until normoglycemic (BGL 5 – 10mmol/L), then as per the treating team or TPN Team. More frequent monitoring may be required depending on the patient's clinical status.
 - If BGL < 4.0mmol/L refer to hypoglycaemia management guideline on the Insulin Subcutaneous Order and Blood Glucose Record – Adult (MR846) chart.
 - If BGL >10mmol/L the treating team should consider subcutaneous insulin.
 - Once normoglycemic (5 to 10 mmol/L) consider reducing BGL monitoring to once or twice daily for non-diabetic patients. For patients with diabetes continue with routine BGL monitoring (QID + 02.00 hours) or as per medical team/TPN Team orders.
- Monitor biochemistry daily, including:¹
 - U&Es
 - Full blood count
 - LFTs
 - Magnesium, phosphate, adjusted calcium.

The treating team is responsible for ordering and monitoring daily pathology for patients on PPN.

The TPN Team is responsible for ordering daily pathology for patients on TPN as they need to be ordered for urgent morning collection to ensure results are available for morning TPN rounds

(Note: The treating team is responsible for any additional bloods required for medical management).

- All patients receiving TPN should have triglyceride levels checked weekly.¹
- Blood should never be taken from the lumen of a central line or peripheral line in which PN is running as the PN solution may potentially contaminate the blood sample, affecting the accuracy of the laboratory results.¹⁵
 - If a central line has two lumens, PN may be paused (do NOT disconnect) for 15 minutes and after flushing the line blood samples may be taken from the secondary lumen. PN should be recommenced after samples are taken.
- Patients receiving longer term TPN (i.e., more than 3 months on the ward or more than 4 weeks in ICU) should have a more comprehensive biochemical nutrition analysis and micronutrient screening, including iron studies, thyroid function, vitamins A, B1, B2, B3, B6, B12, C, D, E, folate, copper, manganese, selenium, zinc, chromium and carnitine. Care should be taken with the interpretation of micronutrient levels in the presence of inflammation due to the acute phase response.¹⁶
- Patients on long-term TPN are at greater risk of gut atrophy, liver dysfunction, hepatic steatosis, and cholestasis. Liver dysfunction, in the absence of sepsis, hepatotoxic medications or underlying liver disease, requires careful review by the TPN Team to ensure the patient is not being overfed and to consider options such as cyclical TPN, fish-oil based lipids, micronutrient deficiencies or small amounts of enteral nutrition to stimulate gall bladder emptying.^{5, 17}

7. Parenteral Nutrition Cessation

- Ideally PN should be titrated with oral or enteral intake as the patient improves so nutritional status is not compromised.¹
- Abrupt cessation of PN when the patient is not receiving any other form of nutrition may lead to hypoglycaemia.⁹ This is less likely with sudden cessation of PPN than TPN but still may occur on rare occasions.
- PN must not be stopped suddenly for diagnostic tests and interventions (except in the case of pausing for blood sampling as outlined above). If the PN infusion is disconnected for any reason, the remainder of the bag must be discarded.
- If the PN is unexpectedly discontinued, access is lost or the PN bag is unavailable, and the patient is not receiving any oral or EN, refer to [Appendix 5: Trouble Shooting Guide](#).
- For patients receiving TPN, the TPN Team will liaise with the treating team to make the decision regarding the continuation or cessation of TPN prior to the next scheduled bag change. For patients receiving PPN, this decision is made by the ward dietitian and / or treating team.

Care should be taken when ceasing PN in patients receiving an insulin infusion. In these cases, the insulin infusion should be ceased prior to ceasing PN and BGLs should be closely monitored.

7.1 To wean TPN:

- Decrease rate by 50% for 4 hours (unless otherwise instructed by the TPN Team).⁹
- Check BGL and cease TPN if BGL is within normal limits.
- Perform BGL one hour following cessation and monitor 6 hourly for 24 hours.
- If BGLs are less than < 4.0mMol/L notify treating team and refer to hypoglycaemia management guideline on the Insulin Subcutaneous Order and Blood Glucose Record – Adult (MR846) chart.

7.2 To wean PPN:

- Due to the lower glucose content of the PPN formulation, it is not essential to taper the rate down prior to cessation.⁹
- Check BGL one hour post cessation and if less than < 4.0mMol/L notify treating team and refer to hypoglycaemia management guideline on the Insulin Subcutaneous Order and Blood Glucose Record – Adult (MR846) chart.

Cease PPN prior to ward transfer if transferring to unit/ward outside of General Surgery (G61, G63, G71), Gastroenterology and Transplant (G63) or ICU.

8. Discharge Planning / Home Parenteral Nutrition

PN in the home environment may be indicated for some patients who are medically stable on their PN solution but have recovered sufficiently from their underlying or precipitating illness. Home parenteral nutrition (HPN) is usually considered for patients requiring medium to long term PN therapy due to intestinal failure that, despite maximal therapy, would otherwise lead to deteriorating nutrition and / or fluid status.¹⁸ Such conditions may include short bowel syndrome, bowel motility disorders leading to recurrent obstruction, gastrointestinal fistulae, or high output stomas. HPN may also be provided as palliative nutrition to patients in the late phases of end stage diseases.⁵

All patients requiring HPN on discharge require an e-referral from the treating team to the TPN Team requesting consideration for HPN. The discharge process for HPN may take up to a minimum of 10 working days to ensure a safe and effective discharge.

8.1 Eligibility Criteria

HPN will only be considered for patients meeting the following criteria:

- The patient is either:
 - a) an inpatient at SCGH and receiving is TPN; or
 - b) resides within SCGH catchment area and requires medium to long term TPN that cannot be managed at an alternative health facility. (Note: in this case, the patient's care will need to be accepted by a treating physician at SCGH and the patient will need to be admitted to SCGH for inpatient HPN training.)
- The patient and / or the patients' legal representative has given their fully informed consent to receive HPN therapy.⁵
- The patient has documented intestinal failure that, despite maximal medical and nutritional therapy, would lead to deteriorating nutrition and / or fluid status.¹⁸
- The patient is sufficiently metabolically stable outside the hospital setting.⁵
- The patient has a consistent PN prescription with adequate BGL control and can tolerate overnight PN administration.
- The patient has appropriate central access for long-term PN.
- The patient and / or carer is physically, mentally and emotionally capable of undertaking HPN training, manage line connections and PN administration in the home environment.
- The patient has a suitable home environment for the safe delivery and administration of HPN⁵, including a separate refrigerator for HPN solution storage.
- The patient is willing to attend outpatient appointments for review as required.

8.1 Preparing the Patient for Discharge

The TPN Team will place a HPN Pre-Discharge Checklist in the front of the patient's medical file and check off tasks as they are completed. The HPN Pre-Discharge Checklist ensures all necessary steps are taken for a seamless discharge and will inform the treating team of the progress of the discharge process.

8.1.1 E-Referral for HPN

Treating team must complete an e-referral to the TPN Team requesting consideration for HPN.

8.1.2 Venous Access for HPN

- The treating team must ensure the patient has appropriate long-term central access with a dedicated line for HPN. The TPN Team or IV Therapy may be consulted for advice if necessary.
- Ideally, the line for HPN should be a right-sided, tunnelled, single lumen catheter specifically placed for the purpose, in consultation with the patient and / or carer to ensure that the exit site is accessible for self-management and so that it may be concealed under clothing.^{5, 17, 18, 19}
- A multilumen catheter may be used if a patient requires other concurrent parenteral therapy but requires dedicated lumens for each different therapy.¹⁸
- Access to the upper vena cava should be the first choice for CVC placement for HPN, via the internal jugular vein or subclavian vein.²⁰ The tip of the CVC should be placed at the level of the right atrial-superior vena cava junction.^{5, 17}
- 6 Fr or 6.6 Fr catheters for HPN are associated with a lower rate of occlusion and are probably less thrombogenic.²¹
- Appropriate CVC options include:¹⁸
 - Tunnelled catheters (e.g., Hickman or Broviac®) – offer acceptable rates of sepsis and thrombosis and ease of use.
 - Fully implanted catheters (e.g., Port-a-cath) – may be better suited for intermittent rather than daily use. Needles to access ports should be replaced at least once per week.⁵
 - PICCs – acceptable for short-term PN (6 – 18 months). PICC extensions may be required to enable self-care.^{5, 18}

- All connections should be luer-locked to prevent accidental disconnection and air embolus.¹⁸
- All external catheters should have a clamp and be locked under positive pressure.¹⁸
- Dressings for the CVC exit site are recommended for the first 10 days. Highly permeable transparent dressings should be changed weekly in the absence of a specific indication and dry gauze should be changed daily. Apart from PICC lines, dry healed exit sites do not require dressings.¹⁸

8.1.3 Baxter Homecare Registration

- HPN compounding is undertaken by Baxter Healthcare in Cannington.
- The TPN Team will complete HPN registration for the patient by completing a Baxter Homecare Patient Registration Form (www.baxterhealthcare.com.au).

8.1.4 Prescription

- The TPN Team is responsible for providing Baxter with an electronic repeating prescription for HPN that is optimally tailored to the clinical and intestinal status of the patient.¹⁸
- Ideally, the HPN infusion should be cyclical to minimise metabolic complications and to maximise quality of life.¹⁸
- The infusion rate should be tapered towards the end of the PN administration to avoid rebound hypoglycaemia.⁵
- If additional bags of IV hydration are required they can be prescribed by the TPN intensivist on the HPN prescription order.
- Prescriptions for all other medications need to be organised by the treating team and ward pharmacist prior to discharge.

8.1.5 Central Line Locks

- Some patients will require central line locks to help prevent infection and extend the life of PN catheter.¹⁹ Central line locks should be prescribed by the TPN intensivist in consultation with IV Access Service and the treating team.
- Taurolock™ (Taurolidine 1%/Sodium Citrate 4% 3mL lock) is an ideal catheter locking solution with anticoagulant and antimicrobial properties which reduce catheter thrombosis and infection.^{5, 17} It should be used as first line therapy.
- Second-line therapy options for locking solutions include:
 - Gentamicin 10mg/sodium citrate 31.3mg in 3mL locks – limited evidence for benefit and risk of development of antimicrobial resistance, especially in patients requiring long term HPN.^{17, 19}
 - Ethanol 70% in water for injection 5mL locks – may be associated with systemic toxicity, catheter occlusion and catheter damage.^{5, 17} Not suitable for polyurethane catheters.¹⁹
 - Heparinised saline 50units in 5ml locks - lacks antimicrobial properties and may promote catheter colonisation and biofilm growth^{5, 17}
- Refer to Non-Tunneled and Tunneled Large Bore Central Venous Catheters (SCGOPHCG) and Infusaport – Implantable Intravenous Delivery System (SCGOPHCG) for further information regarding lock administration and withdrawal.
- The type of lock and volume required to lock lines should be decided prior to discharge and the patient should be trained in their use. The locks should be prescribed by the TPN intensivist on the HPN prescription order for supply by Pharmacy Aseptic Services.

8.1.6 Baseline blood tests

At the commencement of the discharge planning process the TPN Team will organise full HPN nutritional screening including micronutrient levels. These tests include U&Es, LFTs, calcium, magnesium, phosphate, fasting lipids, zinc, selenium, manganese, copper,

carnitine, iron studies, thyroid function tests, INR, and fat-soluble and water-soluble vitamins.

8.1.7 Education

- Patient education regarding the principles of asepsis, safe delivery of HPN, and use of pumps and devices should commence as early as possible and be performed by the ward nursing staff in conjunction with the TPN Team, IV Access Service and other specialties as required.
- HPN Nurse (in conjunction with IV Access Service):
- IV access and care.
 - Awareness of potential CVC related complications, including occlusion, central vein thrombosis, catheter fracture and air embolism or catheter migration. The patient must contact the TPN Team immediately if they are experiencing any difficulty with IV access after discharge.
- Ward nursing staff (in conjunction with HPN nurse):
 - Aseptic principles and hand hygiene.
 - Clean preparation surface.
 - Disinfection of needless connector.
 - Administration of saline flushes.
 - Aseptic PN connection / disconnection.
 - Instillation of line lock.
 - Aspiration of line lock prior to next PN bag.
 - Weekly site care, including catheter hub cleaning.⁵
 - Exit site dressings (if necessary).
 - Urinalysis for glucose (if necessary).²¹
- TPN Team:
 - **Pump Use:** Comprehensive instruction on pump use. Saline bags, TPN lines and filters will be provided by the TPN Team for demonstration / practice purposes. The patient must be made aware of potential mechanical problems that may occur with the line or pump and how to resolve or respond to these issues.¹⁸
 - **HPN Solutions:** Storage and handling of the HPN bags, how to inspect the bags for damage, precipitates or signs of cracking (separation of oil and water components of the solution).^{5, 21}
 - **Sepsis:** Signs of line sepsis and how to respond if feeling unwell or suspect infection. It is essential that the patient to contacts the TPN Team if fever or signs of sepsis occur. Patients must also be instructed to contact their GP or other specialists early if any other illness ensues.
 - **BGLs:** Signs and symptoms of hypo / hyperglycaemia and treatment strategies (although most patients should have stable BGLs prior to discharge and will not require additional monitoring at home). If required, contact the Diabetic Nurse Educator for additional education, provision of a glucometer and registration with NDSS for access to discounted strips and other consumables.
 - **Fluid balance:** Monitoring fluid status, signs and symptoms of over / under hydration and appropriate management.

8.1.8 Additional Discharge Paperwork (to be provided by TPN Team)

- HPN Competency Training Checklist:
 - To be completed by all staff throughout the discharge planning process and filed in the patient's medical record as evidence of demonstrated competence.
 - Nursing staff must be confident that the patient is proficient in the aseptic technique, connection/disconnection of HPN, catheter care and administration of any line locks or additional IV fluids / electrolytes prior to discharge.
 - The TPN Team and patient / carer will confirm competency to self-manage HPN prior to discharge by signing the HPN Competency Training Checklist.

- HPN Patient Contract:
 - Outlines the responsibilities of the patient and the TPN Team to ensure the ongoing supply and safe management of HPN in the home environment. It should be agreed to and signed by the patient / carer and the TPN Team prior to discharge.
 - Upon completion, the patient / carer should be provided a copy of the contract and the original copy should be filed in the patients' medical record.
- HPN Care Plan:
 - The TPN Team will provide the patient / carer with an individualised HPN care plan with contact details for all healthcare providers who have significant roles in their HPN care and a summary of their HPN prescription and delivery arrangements.

8.2 HPN Delivery and Equipment

- Baxter deliver HPN solutions to the patient weekly and the patient / nominated next of kin (NOK) must be home to accept the order and sign for the delivery. Any additional hydration arranged through Baxter will be delivered with the HPN solutions.
- HPN bags will be labelled indicating the composition, date, name, infusion rate and storage instructions.⁵
- Baxter will supply a portable PN infusion pump, HPN lines, retractable IV pole and backpack. Equipment requirements are outlined on the Baxter Homecare Patient Registration Form (www.baxterhealthcare.com.au) which is completed by the TPN Team and faxed to Baxter at the commencement of the discharge planning process. It may take up to 10 working days for the equipment to arrive at the hospital. The TPN Team will deliver all equipment to the patient prior to discharge.
- Pump sets are delivered to the patients home by Baxter monthly (unless otherwise arranged).
- The TPN Team will arrange for Pharmacy Aseptic Services to deliver the patient's pre-drawn central line lock syringes and 10 mL sodium chloride 0.9% Posiflush SP™ syringes weekly and ad hoc additional hydration if required.
- Additional TPN consumables may not be supplied by SCGH (syringes, antiseptic wipes, etc.) and the patient may be required to purchase these through Surgical House (www.surgicalhouse.com.au) or an alternative supplier.

8.3 Discharge Services

- The treating team and nursing staff should work together with the TPN Team and Discharge Co-ordinator to arrange any additional services that may assist the patient with safe and seamless discharge and to provide support with any aspects of PN management at home, including HPN administration, catheter care, dressings and / or regular blood tests. These services may include HomeLink, Complex Needs Coordination Team (CoNeCT) or Silverchain.
- Contact details for the TPN Team and Pharmacy Aseptic Services are provided on the HPN Care Plan for any problems related to HPN administration, formulation, nutritional status, fluid balance or CVC-related complications. The TPN Team can also be contacted for advice if the patient becomes acutely unwell and is unsure whether to present to the Emergency Department.
- Baxter Homecare contact details are provided for any concerns regarding delivery times, pump-related issues, or condition of any of the items delivered to the home.
- The TPN Team will provide a discharge letter to the patient's GP summarising HPN indications, prescription details and contact details for the TPN Team. In most cases, the GP will be responsible any day-to-day medical issues.
- Patients should be encouraged to join Parenteral Nutrition Down Under (PNDU), a self-funded, non-for-profit support group for consumers and carers in Australian and New Zealand on HPN. PNDU provides supplementary HPN education, support and networking

among members, which can lead to improved quality of life (QoL) and depression scores and a reduction in catheter infections.¹⁷

8.4 Monitoring and Follow Up

- HPN patients require routine blood tests to monitor nutritional status. The TPN Team will provide the patient with the required pathology request forms prior to discharge. Alternatively, pathology can be arranged via the patient's GP (with results emailed copied to scqh.pnservice@health.wa.gov.au) according to the schedule requested by the TPN Team on discharge.
- Following initial discharge from hospital, it is suggested bloods be monitored weekly or fortnightly until stabilised and then monitoring can be done less frequently.

Routine drawing of blood samples from CVAD should be avoided if possible due to an increased risk of infections.¹⁴

- Patients on HPN should be seen and assessed on a regular basis by the multidisciplinary TPN Team. The timing will depend on the clinical and nutritional status of the patient and may range from weekly to every six months. Ideally, stable patients on long-term HPN should be seen in the HPN Clinic on a 3 – 6 monthly basis.
- Follow-up appointments should assess the adequacy of the HPN regimen based on clinical, anthropometric and biochemical parameters.¹⁷
- Micronutrient screening should be performed every 6 – 12 months (depending on risk of deficiencies / toxicities). CRP levels should be checked simultaneously with micronutrient levels to assist in the interpretation of results.²²
- Bone metabolism and BMD assessment via DEXA scanning is recommended every 12-18 months.^{5, 21}
- QoL should be regularly measured using validated tools as part of standard clinical care¹⁷.
- Regular re-training of HPN users will ensure patients remain up to date in the latest evidence-based or agreed best practice and protocols in HPN CVAD care²³. This is especially important in patient with repeated CVC-related infections.
- All patients receiving HPN should be monitored for long term PN-related complications, including parenteral nutrition related liver disease (PNALD), renal failure and stones, metabolic bone disease and catheter related complications.

Recommended frequency for nutrition and laboratory monitoring is summarised in [Appendix 6](#).

Compliance, monitoring and evaluation

The Department of General Surgery, Gastroenterology and Intensive Care will be responsible for monitoring compliance with this multidisciplinary guideline document via routine clinical incident review processes and safety and quality reporting mechanisms.

The TPN Team maintain a register all clinical errors associated with inpatient TPN prescription or administration, and PN-related complications arising from patients receiving long-term HPN. Regular safety and quality activities will be completed, including review of SAC events, outcome data, readmissions and other measures of quality (incidence of CRBSIs, QoL, etc).

Related Documents

Legislation:

- Health Practitioner Regulation National Law (WA) Act 2010
- Carer's Recognition Act 2004 (WA).
- Work Health and Safety Act 2020.
- Nursing and Midwifery Board of Australia (NMBA) Code of Professional Conduct for Nurses March 2018
- Western Australia: Medicines and Poisons Act 2014, Medicine and Poisons Regulations 2016, Pharmacy Act 2010

Authorising Policy and Standard/s:

- NMHS Clinical Documentation Policy
- NMHS Infection Prevention and Control Policy
- SCGOPHCG Infection Prevention and Control Manual - Aseptic Technique Policy
- SCGOPHCG Patient Identification Policy
- OPH Procedure Matching Policy
- SCGH Procedure Matching Policy
- Australian Commission on Safety and Quality in Health Care. National Standard for User-applied Labelling of Injectable Medicines, Fluids and Lines. Sydney: ACSQHC, 2015.

Standards, procedures, guidelines:

- CPG021 Non-Tunnelled Central Venous Catheter and Peripherally Inserted Central Catheter
- NPG2 Diabetes Management
- NPG4 Intravenous Therapy
- NPG33 Non-Tunneled and Tunneled Large Bore Central Venous Catheter
- NPG47 Infusaport – Implantable Intravenous Delivery System
- PIVC Guideline: Insertion, Maintenance and Removal
- Refeeding Syndrome Guideline

Supporting documents

- Peripheral Parenteral Nutrition (PPN) Order Form (MR526)
- TPN Prescription Order
- Insulin Subcutaneous Order and Blood Glucose Record – Adult (MR846)
- Intravenous Fluid Therapy Orders (MR410/825)

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Standards Applicable	NSQHS Standards (Version 2): 			

For full details regarding, development, review, consultation, approval, endorsement - contact the Policy Officer for the submission form

Appendix 1: Recommended Parenteral Intake for Adults

Macronutrients^{1, 4, 9, 24}

	Daily Requirements	Additional Comments
Energy	25 – 35kcal/kg	Will depend on age/gender/body composition/disease state/activity levels
Lipid	1 – 2g /kg/day	TPN should contain ~30 – 50% of energy from lipid, PPN may contain 40-60% of energy from lipid. Critically ill patients should not receive more than 1.5g/kg/day of lipid from all sources.
Glucose	4 – 5 g/kg /day Minimum 2g/kg/day	Should not exceed 4-7mg/kg/minute (5-10g/kg/day)
Protein	1.0 – 1.5 g/kg	100g amino acids provides ~ 83g protein substrate and 340kcal
Fluid	25 – 35mL/kg	1500 – 2500mL/day for the average adult

Electrolytes^{5, 9, 25}

	Daily Requirements
Sodium	1 – 2 mmol/kg
Potassium	1 – 2 mmol/kg
Calcium	2.5 – 7.5 mmol
Magnesium	4 – 12mmol
Phosphate	10 – 30mmol
Chloride	As needed to maintain acid-base balance

Micronutrients^{16, 26, 27}

Vitamins	Daily Requirements	Minerals and Trace Elements	Daily Requirements
B1	2.5 -3mg	Chromium	10-15ug
B2	3.6 – 5mg	Copper	0.3-0.5mg
B3	40 – 47mg	Iodine	130ug
B5	15 – 17mg	Iron	1mg
B6	3-4mg	Manganese	55ug
B12	5 – 6ug	Molybdenum	19-25ug
Biotin	60ug	Selenium	60-100ug
Folic Acid	400ug	Zinc	3-5mg
C	110 – 200mg	Fluorine	0-1mg
A	3500IU (1155ug RE)		
D	200IU/5ug		
E	>/=9mg		
K	150ug		

These summary tables provide general recommendations only. Each patient requires individual nutritional assessment by the dietitian and consideration of pre-existing nutritional status, disease state, altered metabolism, additional losses and risk of deficiencies and / or toxicities for specific macronutrient, micronutrient and electrolyte requirements.

Appendix 2: PN Premix Formulation

Premix Type	Amino Acids	Protein Equiv. (g)	Glucose	Lipid	Acetate Chloride (mmol)	Na ⁺ (mmol)	K ⁺ (mmol)	Mg ²⁺ (mmol)	Ca ²⁺ (mmol)	PO ₄ (mmol)	Nitrogen (g)	Total Energy (kJ)	Total Volume (mL)	Rate (mL/hr)	Maximum Total Allowable Electrolytes (mmol per bag)
TPN: A (Olimel N9-840E [®])	400mL (14.2%) 57g	47	400mL (27.5%) 110g	200mL (20%) 40g	54← 45	35	30	4	3.5	15*	9	4472	1000	40	Na ⁺ - 150 K ⁺ - 150 Mg ²⁺ - 5 Ca ²⁺ - 5 PO ₄ ³⁻ - 18*
TPN: B (Olimel N9-840E [®])	600mL (14.2%) 86g	71	600mL (27.5%) 165g	300mL (20%) 60g	81← 68	53	45	6	5	23*	14	6709	1500	60	Na ⁺ - 225 K ⁺ - 225 Mg ²⁺ - 8 Ca ²⁺ - 7 PO ₄ ³⁻ - 27*
TPN: C (Olimel N9-840E [®])	800mL (14.2%) 114g	95	800mL (27.5%) 220g	400mL (20%) 80g	108← 90	70	60	8	7	30*	18	8945	2000	80	Na ⁺ - 300 K ⁺ - 300 Mg ²⁺ - 11 Ca ²⁺ - 10 PO ₄ ³⁻ - 36*
TPN: D (Olimel N12-640E [®])	1070mL (14.2%) 152g	126	530mL (27.5%) 146g	400mL (17.5%) 70g	140← 90	70	60	8	5	29*	24	7942	2000	80	Na ⁺ - 300 K ⁺ - 300 Mg ²⁺ - 11 Ca ²⁺ - 10 PO ₄ ³⁻ - 49*
TPN: E (Olimel N9-840 [®])	800mL (14%) 114g	95	800mL (27.5%) 220g	400mL (20%) 80g	89 90←	0	0	0	0	6*	18	8945	2000	Variable (max 80)	Na ⁺ - 300 K ⁺ - 300 Mg ²⁺ - 11 Ca ²⁺ - 10 PO ₄ ³⁻ - 22*
TPN: F (SMOF Kabiven)	750mL (10%) 75g	62	446mL 187g	281mL 56g	157← 82	60	45	7.5	3.8	19*	12	6700	1477	60	Na ⁺ - 225 K ⁺ - 225 Mg ²⁺ - 7.5 Ca ²⁺ - 7.5 PO ₄ ³⁻ - 22*
TPN: G (SMOF Kabiven)	1000mL (10%) 100g	83	595mL 250g	375mL 75g	209← 70	80	60	10	5	25*	16	9200	1970	80	Na ⁺ - 300 K ⁺ - 300 Mg ²⁺ - 10 Ca ²⁺ - 10 PO ₄ ³⁻ - 30*
PPN: PeriOlimel N4 600E	50.6g	42	150	60	55← 49	42	32	4.4	4	17*	8	5852	2000	80	Nil extra additions

* Includes phosphate from lipid

** all numbers have been rounded

Appendix 3: Custom TPN Formulations / Additives

Macronutrients

Compounding Solution	Unit Size	Provides (per litre)	Energy (per litre)
Amino Acids			
Synthamin 17® (10%)	500mL	100g amino acids (85.5g protein)	1700kJ / 400cal
Vamin 18 ® EF (11%)	500mL	114g amino acids (94.6g protein)	1900kJ / 460cal
Glucose			
Glucose 50%	500mL	500g glucose	8000kJ / 1900cal
Glucose 70%	500mL	700g glucose	11200kJ / 2660cal
Lipid			
ClinOleic ® (20%)	500mL	200g lipid (80% olive oil, 20% soya oil)	8360kJ / 2000cal

Micronutrients

Vitamins	Unit Size	Provides (per unit)
*Soluvit®	10mL	3.1mg thiamine, 3.6mg riboflavin, 40mg niacin, 15mg pantothenic acid, 4mg pyridoxine, 60ug biotin, 400ug folic acid, 5ug cyanocobalamin, 100mg ascorbic acid
*Vitalipid®	10mL	3300IU retinol, 200IU ergocalciferol, 10IU dl-α-tocopherol, 150ug phytomenadione
Cernevit®	5mL	3.51mg thiamine, 4.14mg riboflavin, 46mg niacin, 17.25mg pantothenic acid, 4.53mg pyridoxine, 69ug biotin, 414ug folic acid, 6ug cyanocobalamin, 125mg ascorbic acid, 3500IU retinol, 220IU cholecalciferol, 11.2IU α-tocopherol
Phytomenadione (Vitamin K)	2mg	2000ug phytomenadione
Minerals and Trace Elements	Unit Size	Provides (per unit)
*Addaven®	10mL	10ug chromium, 380ug copper, 1.1mg iron, 55ug manganese, 130ug iodide, 950ug fluoride, 19ug molybdenum, 79ug selenium, 5mg zinc
Zinc Chloride	2mL	5.1mg zinc

*standard inclusion unless otherwise specified.

Other:

Water for injection – as required.

Actrapid® (insulin neutral) – as required.

Appendix 4: B Braun Infusaport Space Pump Programming Guideline

PPN Programming

PPN will be infused at the rate prescribed on the PPN Order Form (MR526).

Procedural Information

1. Select the **Care Unit** using the  and  arrows keys to scroll through the list. Press  to make selection.
2. **New drug Therapy: Use drug library?** Press  to confirm.
3. **Select category** – use  arrow key to scroll to **Fluids/PN/Electrolytes**. Press .
4. Press  to access **PQRS**.
5. Use the  arrow key to scroll down to **Parenteral Nutrition:Peripheral**. Press .

Safety Drug information warning will appear on screen: **ENSURE Dedicated 20 – 22G**

PIVC refer to PN Clinical Practice Guideline. Press .

6. Press  Enter **Rate** in mL/hr. Press  to confirm.
7. Press  Press  Enter **VTBI** in ml. Press  to confirm.
8. Press  Press  Enter **Time** in h:min. Press  to confirm.
9. Press  to initiate infusion.
10. Both nurses must document administration by signing TPN prescription order form / PPN Order from (MR526).

Continuous TPN Programming

PN infusion rates for continuous TPN will be charted by the TPN Team on the TPN Prescription Order.

Procedural Information

1. Select the **Care Unit** using the  and  arrows keys to scroll through the list. Press  to make selection.
 2. **New drug Therapy: Use drug library?** Press  to confirm.
 3. **Select category** – use  arrow key to scroll to **Fluids/PN/Electrolytes**. Press .
 4. Press  to access **PQRS**.
 5. Use the  arrow key to scroll down to **Parenteral Nutrition:Total 24hr**. Press .
- Safety Drug information warning will appear on screen: **USE a CENTRAL line only.**
- Press .
6. Press  Enter **Rate** in mL/hr. Press  to confirm.
 7. Press  Press  Enter **VTBI** in ml. Press  to confirm.
 8. Press  Press  Enter **Time** in h:min. Press  to confirm.
 9. Press  to initiate infusion.

10. Both nurses must document administration by signing TPN prescription order form / PPN Order from (MR526).

Cyclical TPN Programming

TPN infusion rates and duration for patients on cyclical TPN will be charted on the TPN Prescription Order by the TPN Team. Instructions will include:

- Maximum rate
- Ramp duration
- Taper duration
- Volume to be infused (VTBI)

Procedural Information

1. Select the **Care Unit** using the  and  arrow keys to scroll through the list. Press  to make selection.
2. **New drug Therapy: Use drug library?** Press  to confirm.
3. **Select category** – use  arrow key to scroll to **Fluids/PN/Electrolytes**. Press .
4. Press  to access **PQRS**.
5. Use the  arrow key to scroll down to **Parental Nutrition:Overnight**. Press 
Safety Drug information warning will appear on screen: **Overnight TPN regime. USE a CENTRAL line only**. Press .
6. Press  Enter **Maximum** in mL/hr. Press  to confirm.
7. Press  Press  Enter **Ramp** in h:min. Press  to confirm
8. Taper time pre-programmed as 2 hours. Visualise to confirm.
9. Press  Press  Enter **VTBI** in ml. Press  to confirm.
10. Press  to initiate infusion.
11. Both nurses must document administration by signing TPN prescription order form.

Appendix 5: Trouble Shooting Guide

Problem	Action
After hours TPN referral	Complete e-referral and contact ICU to notify them of the referral. PN rounds still occur on the weekend and are run by the ICU Consultants.
After hours PPN referral	Referrals received after 15.00 hours will be commenced the following day. PPN may be prescribed by the treating team if the dietitian is unavailable, and the patient should be reviewed by the dietitian as soon as possible.
TPN bag hasn't arrived on the ward	If the patient has recently transferred from another ward, check the TPN hasn't been delivered to the previous ward. Check script tracker to confirm if PN was made and to which ward it was allocated. If unable to locate PN from above sources, contact TPN Team/on-call pharmacist/ICU to arrange a standard TPN TCB and in-line filter from ICU stock.
PPN bag hasn't arrived on the ward	Contact the treating team to see if alternative therapy (i.e. IV hydration or dextrose infusion) is required until the next PPN bag is available at 18.00 hours the following day.
Standard TPN bag from ICU stock is required	For a standard TPN TCB from ICU stock (Olimel® N9-840E 1000mL) that does not have an accompanying patient prescription, the PN must be charted on the Intravenous Fluid Therapy Orders (MR410check/825) chart.
PN bag is damaged or leaking, contains visible precipitates	DO NOT USE. Notify treating team for PPN or TPN Team for TPN. If current TPN bag runs out commence 10% Dextrose at the same rate as the TPN infusion until the new TPN bag is available. Monitor BGLs.
Blocked Central Access	Run 10% dextrose via a peripheral cannula at same rate as TPN and monitor BGLs. If unable to unblock the line, contact IV Access Service for assistance within working hours. Notify TPN Team.
TPN suddenly runs out or is disconnected	Flush lumen and discard the remaining TPN solution. Do not reconnect the TPN. Treating team to chart 10% Dextrose at the same rate as the TPN infusion until the new TPN bag is available. Notify the TPN team and monitor BGLs.
PPN suddenly runs out or is disconnected	Flush cannula and discard the remaining PPN solution. Do not reconnect the PPN. Check BGL one hour post cessation. Inform the treating team.
Patient needs to cease unexpectedly TPN	Notify the TPN Team. It is recommended that the PN rate be halved for four hours prior to cessation to prevent rebound hypoglycaemia. Check BGL 1 hour post TPN cessation.
Patient needs to cease unexpectedly PPN	Cease infusion and flush cannula. Check BGL one hour post cessation. Inform the treating team.

For any other issues concerning PPN: Contact the ward dietitian or ward pharmacist.

For any other issues concerning TPN: Contact the TPN Team or ICU.

Appendix 6: HPN Nutrition Monitoring

Nutrition and Laboratory Monitoring Guidelines for HPN Patients ⁵	
MEASURE	FREQUENCY
Weight	At baseline then weekly. Patient must contact HPN Service if experiencing ongoing weight loss.
Body composition	At baseline, then 3 - 6 monthly.
Fluid balance	Patient to monitor at home if experiencing excessive losses via vomiting, diarrhoea, stoma/fistula output, sweating, etc. or pitting oedema/leg swelling etc.
Glucose	Patient to check levels if they are feeling unwell and have diabetes, a history of unstable sugars on PN, or large amounts of insulin in their PN formulations.
Temperature	Patient to check at home if feeling unwell.
Catheter exit site	Daily
U&E	At baseline. Fortnightly for the first 2 months, then monthly for the next 4 months. Every 3 - 6 months if stable
Calcium, magnesium, phosphate	At baseline. Fortnightly for the first 2 months, then monthly for the next 4 months. Every 3 - 6 months if stable
LFTs (including INR)	At baseline. Monthly for the first 6 months. Every 3 - 6 months if stable.
FBP	At baseline. Fortnightly for the first 2 months, then monthly for the next 4 months. Every 3 - 6 months if stable
TG and cholesterol	At baseline. Monthly for the first 6 months. Every 6 months if stable.
Zinc	At baseline. As required or every 6 months if stable
Selenium	At baseline, then 6 monthly.
Manganese	At baseline, then 6 - 12 monthly.
Copper	At baseline, then 6 monthly.
Iron studies	At baseline, then 6 monthly.
Thyroid function	At baseline, then 6 monthly.
Molybdenum	At baseline, then 6 monthly. NOTE: No test available for Mb in Australia)
Vitamin A, D, E	At baseline, then 6 monthly.
Vitamin B1, B2, B3, B6, B12, C, Folate	At baseline, then 6 monthly.
Chromium	Only if deficiency is suspected
Biotin	Only if deficiency is suspected
Carnitine	At baseline, then 6 monthly.
Aluminium	Only if toxicity is suspected.
Bone mineral density screening	Every 12-18 months.
Liver ultrasound	Every 12 months.

Adapted from ESPEN guideline on home parenteral nutrition. Clin Nutr. 2020;39(6): 1645 - 1666